

Sensor-Driven Digital Twin for Collision Prevention in Autonomous Systems

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Autonomous systems require reliable perception to operate safely in complex environments where sensor limitations can increase collision risk. This research investigates a multi-sensor fusion framework supported by a digital twin approach to improve robustness and safety.

- Integration of camera, LiDAR, and mmWave radar
- Real-time processing on embedded hardware
- Time-to-Collision estimation for risk assessment
- Simulation-based evaluation of system behaviour
- Scalable design for autonomous platforms

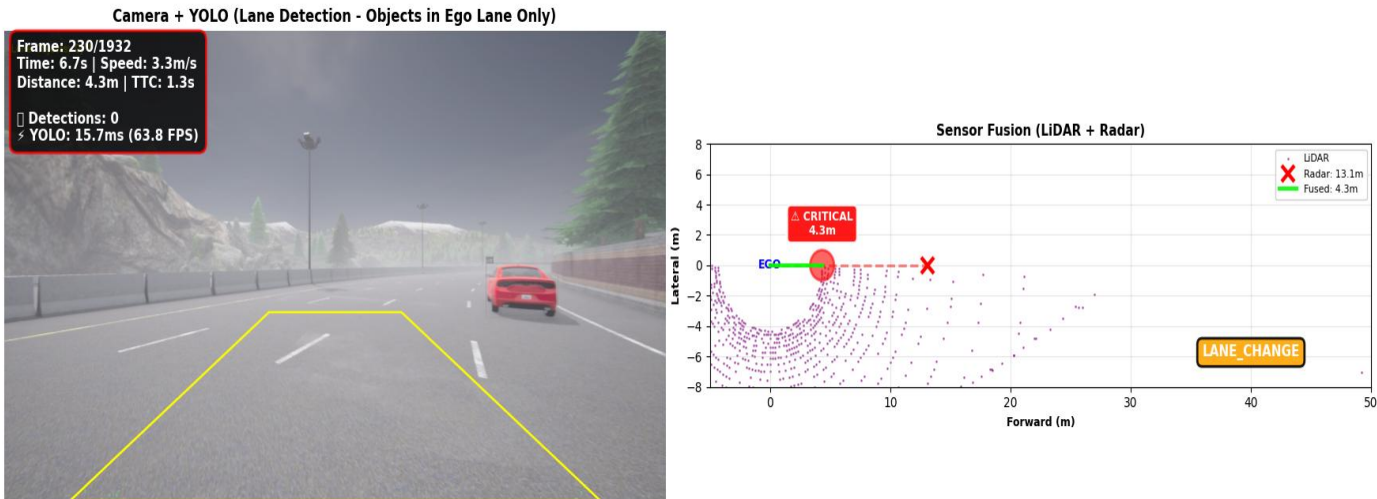


Image 1: Clear Conditions Sensor fusion Driving Simulation

Multi-sensor fusion and digital twin research for improving perception reliability and collision prevention in autonomous systems

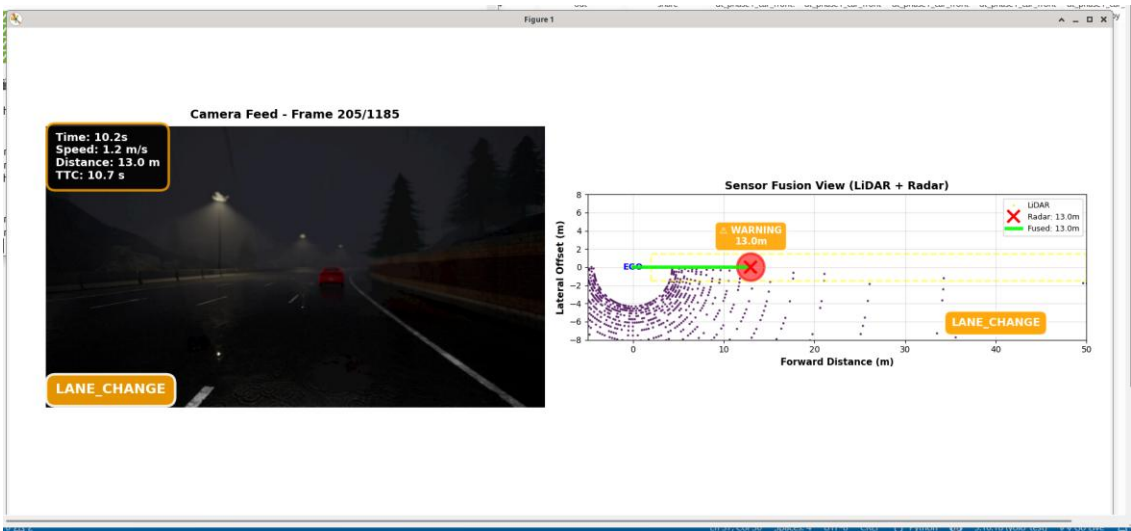


Image 2: Dark Rainy Conditions Sensor fusion Driving Simulation

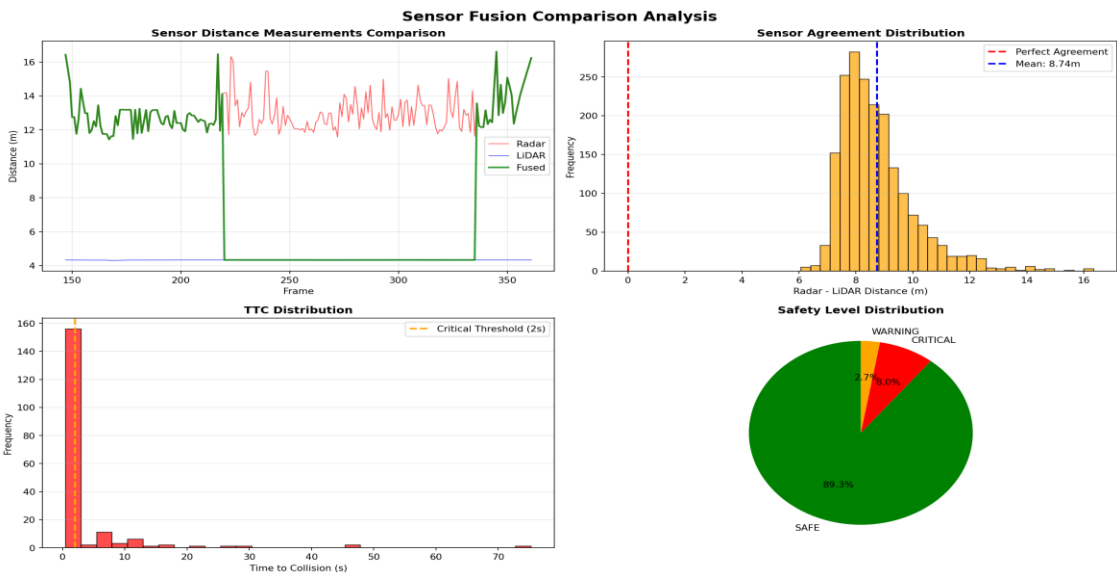


Image 3: Sensor Fusion Analysis

Autonomous systems operate in safety-critical environments where perception failures can lead to severe consequences. Existing approaches that rely on a single sensing modality are vulnerable to environmental factors such as low visibility, adverse weather, and occlusion, resulting in reduced reliability and increased collision risk. This research investigates a sensor-driven digital twin framework that integrates camera-based computer vision, LiDAR distance sensing, and mmWave radar velocity measurement to improve perception robustness. A deep learning model based on the YOLO architecture is used to detect and classify objects from visual data, while LiDAR provides accurate spatial information and radar supplies motion and velocity measurements. These complementary sensor outputs are fused to estimate Time-to-Collision, enabling early identification of potential hazards and supporting collision-prevention strategies. The proposed framework is evaluated using the CARLA autonomous driving simulator to analyse system performance under controlled yet realistic conditions. The research aims to contribute to safer and more reliable autonomous technologies through improved multi-sensor perception and validation methods.

References

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